

□ BIOLOGY CH.4 — DEFICIENCY & DISEASES — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Medicinal Plants — Sarpagandha & Arjuna

- 'Sarpagandha' blood pressure treat karne ka best remedy hai — diabetes, insomnia, indigestion mein bhi use hota hai. Rajnigandha stress reliever hai, Navamallika cardiac tonic hai. [Q107]
- 'Arjuna' plant (Terminalia arjuna) high blood pressure aur earache treat karne mein use hota hai — leaves blood pressure regulate karti hain, juice earache cure karta hai, bark heart diseases cure karti hai. [Q109]

2. HIV/AIDS — Transmission, Prevention & Testing

- AIDS (Acquired Immune Deficiency Syndrome) HIV (Human Immunodeficiency Virus) se hoti hai — white blood cells attack karta hai. ELISA test se antibodies detect hoti hain, World AIDS Day 1 December hai. [Q108]
- ⚠️ Trap: HIV infection prevent karne ke liye 'Available vaccines' use karna sahi tareeka NAHI hai — koi HIV vaccine exist nahi karti. Safe sex, sterile needles, aur safe blood transfusion effective preventive methods hain. [Q110]
- ⚠️ Trap: 'HIV patient ke saath toilet share karna' se HIV NAHI failta — HIV sexual contact, blood transfusion, mother-to-child (pregnancy), ya syringes share karne se failta hai. [Q114]
- Condoms sexually transmitted diseases (STIs, HIV samet) ke spread ko rokne mein effective contraception hain. [Q115]
- HIV avoid karne ki best strategy hai 'sex ko ek known-healthy person tak restrict karna' (monogamy). [Q119]
- ⚠️ Trap: HIV-AIDS virus 'hand shaking' se transmit NAHI hota — blood transfusion, sexual contact, aur breastfeeding se failta hai (body fluids ke exchange se). [Q120]
- Person 1 saal pehle HIV-infected hai ya nahi, ye 'anti-HIV antibodies blood mein' dhoondkar diagnose kiya jaata hai — 3 types ke HIV tests: Antibody, Antigen, Nucleic acid tests (NAT). [Q122]
- HIV transmission prevent karne ke liye 'Barrier methods only' (jaise condoms) effective hain — Copper-T ya oral contraceptives HIV transmission nahi rokta, sirf pregnancy prevent karte hain. [Q123]
- HIV ka full form 'Human Immunodeficiency Virus' hai — ye immune system attack karta hai (specifically CD4 T lymphocytes), body ki infections se ladne ki capacity kamzor karta hai. [Q171]

3. UV Rays — Skin Cancer

- Excessive UV rays exposure se 'Skin cancer' hota hai — immune system damage aur Cataracts (eye damage) bhi ho sakti hai. Leukemia blood cancer hai (WBC count badhne se). [Q111]

4. Puberty — Voice Change

- Puberty ke dauran voice change 'larynx ke development' se hoti hai — boys mein testosterone badhne se larynx mein changes aate hain. [Q112]

5. Tooth Decay

- ⚠️ Trap: Tooth decay mein 'bacteria ka koi role nahi, virus se hoti hai' — ye statement GALAT hai — actually bacteria plaque mein sugars ko acids mein convert karta hai, jo enamel soften karke tooth loss karti hai. [Q113]

6. STDs — Virus vs Bacteria vs Fungus

- STDs jo virus se failte hain: AIDS (HIV), Warts (HPV), Genital Herpes (Herpes Simplex Virus) — ⚠️ Gonorrhoea aur Syphilis bacteria se failte hain, isliye inhe confuse na karein. [Q116, Q117]
- 'Warts' HPV (human papillomavirus) se hote hain — excess keratin protein produce karta hai. [Q118]
- ⚠️ Trap: 'Candidiasis' sexually transmitted disease NAHI hai — ye Candida yeast se hone wala fungal infection hai. Syphilis, Gonorrhea aur Warts actual STDs hain. [Q169]

7. Vitamin K — Blood Clotting

- Vitamin K (Phylloquinone) proteins-calcium activate karta hai blood clotting ke liye — green leafy vegetables (kale, spinach, broccoli) mein milta hai. Deficiency se poor bone development, Osteoporosis hoti hai. [Q131]

8. Dengvaxia — First Approved Dengue Vaccine

- US FDA ne 2019 mein 'Dengvaxia' vaccine ko Dengue disease ke liye approve kiya — 9-16 saal ke logo mein, jo pehle dengue-infected the aur endemic areas mein rehte hain. [Q132]

9. Kala-azar — Caused by Leishmania

- Kala-azar 'Leishmania' unicellular protozoan parasite se hoti hai — infected female phlebotomine sandflies ke bite se failti hai. Symptoms: Fever, weight loss, spleen-liver enlargement. [Q133]
- Kala-azar organism ki origin 'downward gamete fission' (binary fission) se hoti hai — India mein 'Leishmania donovani' hi is disease ka ekmatra parasite hai. [Q167]

10. Bird Flu — H5N1 Virus

- Bird Flu (Avian Influenza) 'H5N1 virus' se hota hai — ek subtype of Avian influenza Type A viruses. [Q134]

11. Vitamin A — Vision & Its Deficiency

- Vitamin A ki deficiency se color blindness aur poor vision hoti hai — night blindness bhi hoti hai. [Q135]
- Carrots mein high Vitamin A hoti hai (beta carotene se) — bachche ki vision strengthen karti hai. [Q124]
- Vitamin A eyes-skin ko healthy rakhti hai — immune system-reproduction ke liye bhi important hai. [Q145]
- 'Golden rice' Vitamin A ki deficiency fulfill karti hai — Vitamin A deficiency se Night blindness hoti hai. [Q160]

12. Eye Diseases — Goiter is NOT One

- ⚠️ Trap: 'Goiter' eye disease NAHI hai — ye thyroid gland ka abnormal enlargement hai (iodine deficiency se). Glaucoma, Cataract aur Dry eye actual eye conditions hain. [Q136]

13. Disease-Causative Organism — A Match Trap

- ⚠️ Trap: 'Cholera-Influenza' galat match hai — Cholera actually *Vibrio cholerae* bacterium se hoti hai, Influenza se NAHI. (TB-*Mycobacterium*, AIDS-HIV, Malaria-*Plasmodium* sahi hain.) [Q137]

14. Fungal Plant Diseases

- *Puccinia graminis* fungus 'Black rust in wheat' cause karti hai — spores wind-rain se spread hoti hain. (Late blight of potato — *Phytophthora infestans*; Red rot of sugarcane — *Colletotrichum falcatum*.) [Q138]
- '*Gibberella fujikuroi*' fungal pathogen rice seedlings mein 'bakane' (foolish seedling) disease cause karta hai. [Q142]
- ⚠️ Trap: 'Citrus canker' fungal disease NAHI hai — ye bacterial disease hai (*Xanthomonas citri*). Damping off seedling, Rust of wheat (*Puccinia*) aur Red Rot of sugarcane fungal diseases hain. [Q146]
- ⚠️ Trap: 'Potato mosaic' seed-borne disease NAHI hai — ye aphids ya mechanical contact se failti hai. Moulting of wheat, Red rot of sugarcane, Sheath rot on rice seed-borne diseases hain. [Q154]

15. Malaria — Plasmodium

- '*Plasmodium*' malaria parasite hai — *E.coli*, *Salmonella* aur *Mycobacterium* doosre bacterial pathogens hain. [Q139]

16. Radioisotope — Strontium-89 for Bone Cancer

- 'Strontium-89' metastatic bone cancer ke treatment ke liye use hota hai. Carbon-14 archaeological dating mein, Iodine-131 thyroid diseases mein use hota hai. [Q140]

17. Megaloblastic Anaemia — Vitamin B9 Deficiency

- Megaloblastic anaemia Vitamin B9 (Folic acid) ki deficiency se hoti hai — red blood cells normal se bade ho jaate hain. [Q141]

18. Tranquilizer — Anxiety Treatment

- Tranquilizer normally 'Anxiety' (fear, dread, uneasiness) ke liye administer hoti hai. [Q143]

19. Bacterial Diseases — Diphtheria, Typhoid & Tetanus

- 'Diphtheria' bacteria (*Corynebacterium diphtheriae*) se hoti hai — throat-upper respiratory tract affect karti hai. Chicken pox, Mumps, Measles viral diseases hain. [Q144]
- 'Typhoid' bacteria (*Salmonella Typhi*) se hoti hai. Polio, Measles, Hepatitis A viral diseases hain. [Q129]
- Tetanus (lockjaw) nervous system ki serious disease hai, toxin-producing 'Bacterium' (*Clostridium tetani*) se — muscle contractions (jaw-neck) karti hai. [Q126]

20. Chicken Pox — A Viral Disease

- Chicken Pox 'virus' (Varicella Zoster Virus) se hoti hai — highly contagious. Virus diseases: Rabies, Measles, Herpes, Hepatitis, Polio, AIDS, Smallpox, Dengue. Bacteria diseases: plague, diphtheria, typhoid, tetanus, cholera. [Q127]

21. Fire Blight — Apple's Disease

- Apple se related disease 'Fire blight' hai — highly contagious bacterial infection, fruit trees (apple, pear) ko affect karti hai, wilting-blackening karti hai. Tikka disease groundnut ki, green ear disease millet ki hai. [Q128]

22. Hepatitis A — A Viral Disease

- Hepatitis A 'virus' (hepatitis A virus/HAV) se hoti hai — liver ko affect karti hai, contaminated food-water ya close contact se transmit hoti hai. [Q130]

23. Peptic Ulcer — *Helicobacter pylori*

- Peptic ulcer ka causative organism 'Bacterium' hai — *Helicobacter pylori*. Protozoa diseases: Malaria, Leishmaniasis, Toxoplasmosis. [Q147]

24. Airborne vs Waterborne Diseases

- ⚠️ Trap: 'Cholera' hawa se NAHI failti — Tuberculosis, Pneumonia aur Common cold airborne diseases hain, Cholera contaminated food-water se failti hai. [Q148]

- ⚠️ Trap: 'Plague' water pollution se NAHI hoti — ye bacterium (*Yersinia pestis*) se, flea bites ke through failti hai. Cholera, Diarrhoea aur Typhoid water pollution se related hain. [Q152]

25. Filariasis (Elephantiasis) — Mosquito-Borne

- 'Filariasis' (Elephantiasis) mosquito-borne disease hai — nematode parasites (Filarioidea family) infected mosquito bite se failte hain. [Q149]
- 'Elephantiasis' se legs mein excess swelling hoti hai — Lymphatic filariasis, nematode parasites se, mosquito bites se failti hai. [Q156]
- 'Filarial worms' (*Wuchereria*) elephantiasis cause karte hain — lymphatic vessels mein chronic inflammation hoti hai. [Q170]

26. Eye Donation — Who Can Donate

- 'Diabetes' patients eyes donate kar sakte hain — hypertension-asthma patients bhi. AIDS, Hepatitis B/C, rabies, tetanus, cholera, leukemia wale nahi kar sakte. Eyes 6-8 ghante ke andar donate honi chahiye. [Q150]

27. Antibiotics — Stop Microorganism Growth

- 'Antibiotics' medicines disease-causing microorganisms ki growth rokti hain — bacterial infections fight karti hain (Amoxicillin, Ciprofloxacin jaisi). Antibodies immune system ke proteins hain, Analgesics pain relieve karte hain. [Q151]

28. Amoebiasis — Causes Dysentery

- 'Amoebiasis' Dysentery cause karti hai — *Entamoeba histolytica* parasite se, large intestine mein infection, kabhi-kabhi liver-lungs-brain tak spread ho sakti hai. [Q153]

29. Foot-and-Mouth Disease — A Viral Illness

- Cattle mein Foot-and-mouth diseases 'Virus' se hoti hain — highly contagious. Animal viral diseases: Bird Flu, Rabies, Rinderpest. [Q155]
- 'Ephepi epizootica' (disease of feet and mouth) 'Virus' se hoti hai. [Q165]

30. Leukemia — Blood Cancer

- 'Leukemia' blood cells (bone marrow samet) ko affect karti hai — DNA mutation se hoti hai. Polio virus se, Meningitis bacteria-virus-fungi se, Smallpox variola virus se hoti hai. [Q157]

31. Leukoderma — A Skin Disease

- 'Leukoderma' skin ki disease hai — skin pigmentation (melanin) loss hoti hai, skin whitening hoti hai (hands, feet, face affect hote hain). *[Q158]*

32. Typhoid — Incubation Period

- Typhoid bacillus (Salmonella Typhi) ka incubation period '8 to 14 days' hota hai — Widal test se confirm hoti hai, vaccine available hai. *[Q159]*

33. Fluoride — Tooth Decay Prevention

- 'Fluoride' water mein add ki jaati hai tooth decay prevent karne ke liye — tooth enamel (hydroxyapatite) ko protect karti hai bacteria-produced acid se. *[Q161]*

34. Anaemia — Red Blood Corpuscles Deficiency

- Anaemia 'Red Blood Corpuscles' (RBC) ki deficiency se hota hai — RBC bone marrow mein banti hai, hemoglobin oxygen carry karti hai. *[Q162]*

35. Diabetes — Treated by Insulin

- 'Diabetes' Insulin se treat hoti hai — pancreas enough insulin nahi banata ya body effectively use nahi kar paati. Types: Type 1, Type 2, Gestational. *[Q163]*

36. Grover's Disease — A Skin Disorder

- 'Grover's disease' ek ventilated illness hai — skin disorder, small raised pink itchy spots ka rash hota hai. *[Q164]*

37. Skin Pigmentation Diseases — Vitiligo & Albinism

- 'Sickle cell anemia (Vitiligo)' skin pigmentation development ko affect karti hai — skin ka ek part pigment lose kar deta hai. Albinism mein Melanin poori tarah/partially absent hoti hai. *[Q166]*

38. Goitre — Iodine Deficiency

- 'Goitre' iodine deficiency se hoti hai — thyroid gland abnormal enlarge ho jaati hai. Iodine sources: Seafood, Seaweed, Iodised salt. [Q168]

39. Haemophilia — A Blood Clotting Disorder

- 'Haemophilia' un genes ki absence se hoti hai jo blood clotting factors produce karte hain — sex-linked recessive hereditary disorder hai, minor cut se non-stop bleeding hoti hai. [Q172]

40. Beriberi — Vitamin B1 Deficiency

- Beriberi ka symptom 'weak muscles aur low energy' hai — body mein enough thiamine (Vitamin B1) nahi hoti, jisse walking difficulty, sensation loss, muscle paralysis hoti hai. [Q121]

41. Dengue — Caused by Viruses

- Dengue 'Viruses' se hoti hai — infected mosquitoes (mainly Aedes aegypti, kabhi Aedes albopictus) ke bite se humans mein transmit hoti hai. [Q125]